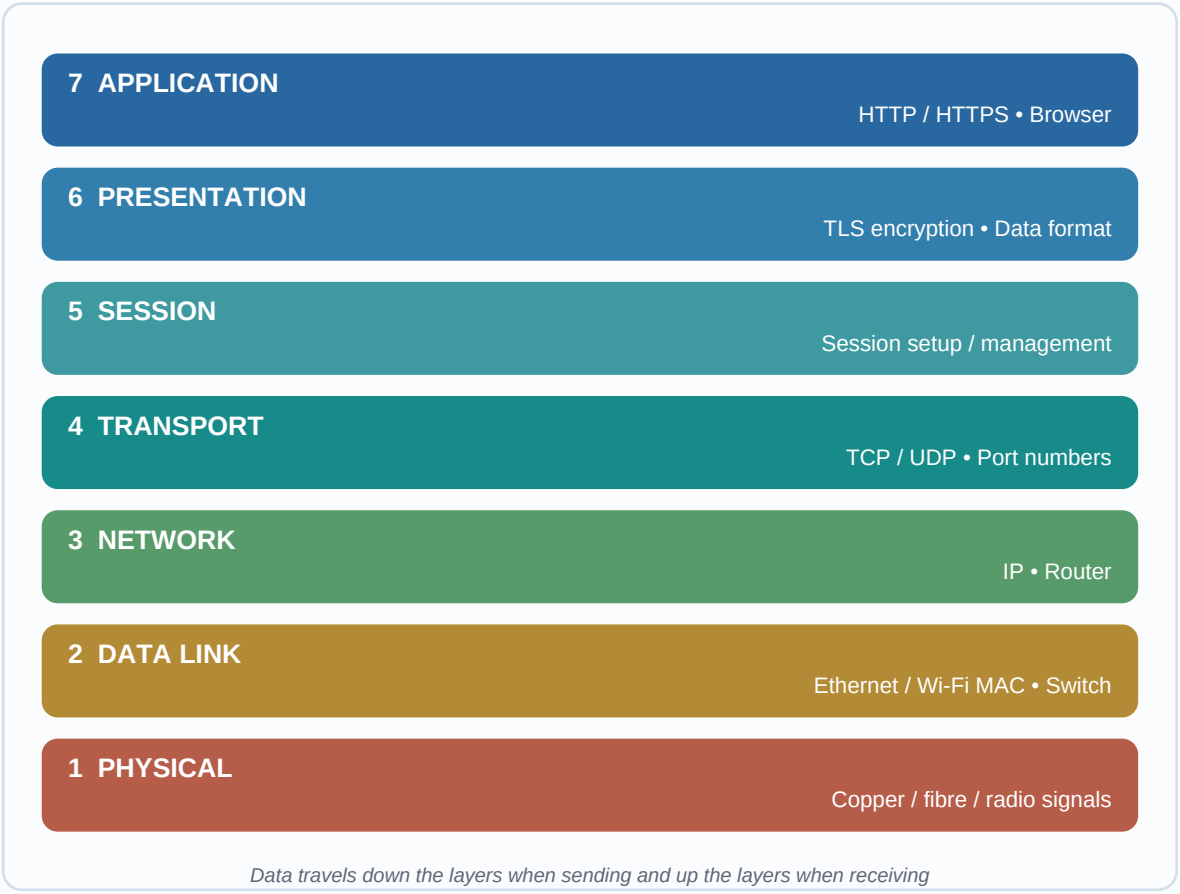
TASK 1 — Draw and Label the OSI 7-Layer Model

The OSI (Open Systems Interconnection) model divides network communication into seven layers. Each layer has a particular role and works with the layers directly above and below it. The model is useful for learning networking concepts and troubleshooting problems systematically.



Data travels down the layers when sending and up the layers when receiving

Layer names and examples

No.	Layer	Main responsibility	Example
7	**Application**	Network services used by applications	HTTP/HTTPS, DNS, browser
6	**Presentation**	Data format, encoding, encryption	TLS encryption, UTF-8
5	**Session**	Starts, manages and ends communication sessions	Session management / RPC concepts
4	**Transport**	End-to-end delivery, ports, segmentation	TCP, UDP
3	**Network**	Logical addressing and routing	IP, router
2	**Data Link**	Frames, MAC addressing, local-link error detection	Ethernet, Wi-Fi MAC, switch
1	**Physical**	Transmits raw bits as signals	Copper cable, fibre, radio

TASK 2 — Match Scenarios to OSI Layers

For each scenario, identify the layer that is most directly responsible. Some real network functions involve more than one layer, but the answers below use the standard introductory OSI mapping.

Answers

a) Encrypting data before sending

Answer: Presentation Layer (Layer 6)

The Presentation layer is commonly associated with encryption, decryption, encoding and data-format translation in the OSI teaching model. In modern networks, TLS encryption is usually implemented through software libraries and does not map perfectly to a single OSI layer.

b) Routing packets across networks

Answer: Network Layer (Layer 3)

Routers use IP addresses and routing tables to forward packets between different networks.

c) Converting data to electrical signals

Answer: Physical Layer (Layer 1)

The Physical layer sends bits as electrical, optical or radio signals over a medium such as copper cable, fibre or Wi-Fi radio.

d) Opening a web page in your browser

Answer: Application Layer (Layer 7)

The browser uses application-layer protocols such as HTTP or HTTPS to request web content. Other layers carry the data underneath.

e) Detecting and correcting errors in frames

Answer: Data Link Layer (Layer 2)

The Data Link layer works with frames and commonly detects link-level errors using checks such as an Ethernet frame check sequence (CRC). Some link technologies also support retransmission or correction.

Answer format requested

a) Encrypting data before sending: Presentation Layer (Layer 6)
b) Routing packets across networks: Network Layer (Layer 3)
c) Converting data to electrical signals: Physical Layer (Layer 1)
d) Opening a web page in your browser: Application Layer (Layer 7)
e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

TASK 3 — OSI Layers in a Daily App

Example app: WhatsApp. When a user sends a message or receives a photo, multiple networking layers cooperate. The following is a simplified learning example; the exact implementation details can vary by app version and platform.

What happens when a message is sent?

1. Application Layer (Layer 7)

WhatsApp creates the message or media request and uses application-level logic to send it to the service. The user interacts with the chat screen, and the app manages message-related information.

2. Presentation Layer (Layer 6)

Data must be represented in a format the communicating systems understand. Encryption is also important for private messaging; in the simplified OSI model, encryption is commonly associated with the Presentation layer, though real implementations can cross layer boundaries.

3. Transport Layer (Layer 4)

Transport protocols carry data between endpoints. Depending on the service and platform, communication may use TCP-based connections or other transport mechanisms. Transport handles functions such as ports and delivery between processes.

4. Network Layer (Layer 3)

IP addressing and routing help packets travel from the phone across the mobile or Wi-Fi network and internet toward the destination service.

5. Data Link Layer (Layer 2)

On the local link, Wi-Fi or cellular link technology carries frames between the device and nearby network equipment. Link-layer mechanisms can detect certain transmission errors.

6. Physical Layer (Layer 1)

The bits are transmitted over radio waves between the phone and Wi-Fi access point or cellular base station, and over electrical or optical signals on wired network segments.

Receiving a message

The process broadly runs in reverse: signals are received, frames and packets are processed, transport data is delivered to the correct app process, and WhatsApp displays the message. The app may also verify and decrypt protected content before showing it to the user.

Why this matters

If Wi-Fi disconnects, the problem may be near the Physical or Data Link layer. If the IP route fails, the Network layer may be involved. If a secure connection or application request fails, Transport, security-related functions or the Application layer may need investigation.

TASK 4 — Website Fails, but Ping Works

Scenario: You cannot open a website in a browser, but you can successfully ping the website's IP address. This indicates that IP connectivity to that address is at least partly working, but it does not prove that every service or network path is healthy.

Most likely layers to investigate

Application Layer (Layer 7)

The website server or web application may be down, misconfigured, returning errors, or unable to serve the requested page.

Transport Layer (Layer 4)

Ping uses ICMP, not TCP. A successful ping does not prove that the web server's TCP port 80 (HTTP) or 443 (HTTPS) is reachable. A firewall may allow ICMP while blocking web traffic, or the web service may not be listening.

Presentation / security functions (often associated with Layer 6)

HTTPS may fail because of certificate errors, TLS version or cipher incompatibility, incorrect system date/time, or a security inspection device. These functions do not always map neatly to one OSI layer in real implementations.

Application name resolution (Layer 7; DNS)

If the website is accessed by domain name, DNS may be failing or returning an incorrect address even though a ping to the IP address works. Test the domain name separately.

Why the Network layer is less likely—but not completely ruled out

A successful ping to the same IP suggests that the host is reachable using ICMP and that at least one IP path works. However, web traffic may use a different route, be filtered differently, or be affected by MTU or routing policies. Therefore, Layer 3 problems are less obvious, not impossible.

Suggested troubleshooting order

1. Try opening the site by IP address and by domain name (where supported).
2. Check DNS resolution for the domain.
3. Test whether TCP port 443 or 80 is reachable.
4. Check browser proxy settings, firewall rules and VPN configuration.
5. Inspect HTTPS certificate/TLS errors and the web server status.
6. Compare access from another device or network.

Conclusion

The most likely causes are at the **Application layer (Layer 7)** or **Transport layer (Layer 4)**, with DNS and HTTPS/TLS configuration also worth checking. Ping uses ICMP, so it confirms only limited IP reachability—not that HTTP or HTTPS is working.

EXTRA PAGE — OSI Troubleshooting Quick Reference

The OSI model helps narrow down a fault by starting with the symptom and checking each layer in a logical order. In practice, protocols and functions do not always fit perfectly into a single layer, but the model is a useful study and troubleshooting framework.

Quick reference table

Layer	Typical issue	Example check
7 — Application	Web app error, wrong URL, service unavailable	Check URL, server status, HTTP response
6 — Presentation	Encryption, certificate or data-format issue	Inspect TLS/certificate and data format
5 — Session	Session not established or unexpectedly closed	Check session logs and connection state
4 — Transport	Port blocked, TCP connection fails	Test TCP port; inspect firewall rules
3 — Network	IP address, route or gateway issue	Check IP configuration, route and ping
2 — Data Link	Wi-Fi association, switch port or frame issue	Check link lights, Wi-Fi association, interface
1 — Physical	Broken cable, no signal, radio interference	Check cable, power, signal and link status

Memory trick

From Layer 7 down to Layer 1: **Application, Presentation, Session, Transport, Network, Data Link, Physical.** One common mnemonic is “All People Seem To Need Data Processing.”

Final summary

The OSI model provides seven conceptual layers for understanding network communication. For the website scenario, a successful ping does not guarantee that web browsing will work: DNS, TCP ports, HTTPS/TLS and the web application itself must also be checked.